PATHOPHYSIOLOGY — NURSING

Hematopoietic Function

NRSG 224 — Blood / Recorded Lecture 3

Complete exam-detail notes from the PowerPoint lecture recording. Slide content plus what the instructor said out loud — definitions, numbers, mechanisms, clinical applications, and every comparison she flagged.

1. Hematopoietic function — the big picture

Hematopoietic function = how the body makes blood cells. This lecture walks the three cell lines that come from bone marrow stem cells, then what goes wrong when you have too many, too few, or poorly working cells in each line.

She said (not on the slide): *She opened by asking: what three kinds of cells does bone marrow make? Answer them by BOTH names.*

Line	Formal name	Everyday name	Main job
Red cells	Erythrocytes	RBCs	Carry O2 (via hemoglobin / iron)
White cells	Leukocytes	WBCs	Defend against infection / foreign material
Platelets	Thrombocytes	PLTs	Primary hemostasis — platelet plug

TEST TIP: If a question asks for the three products of bone marrow, give both names. She used erythrocytes / leukocytes / thrombocytes on purpose.

2. Hematopoiesis and differentiation

All three lines come from hematopoietic stem cells in red bone marrow. New cells pass through levels of differentiation until they become mature circulating cells.

- Process name: hematopoiesis.
- Regulated by growth factors — she named erythropoietin (EPO) and granulocyte colony-stimulating factor (G-CSF).

She said (not on the slide): *The reason she mentioned growth factors is clinical: we can now make synthetic versions in the lab and you will give them. Synthetic EPO-type drugs for anemias; G-CSF when neutrophils are too low.*

She then said: focus on red cells first. The rest of the lecture follows RBC → hemostasis/platelets/factors → WBC → leukemia (WBC cancer).

3. Red cells, hemoglobin, and erythropoiesis

Job of the RBC

Transport oxygen to tissues. That works because of iron inside hemoglobin. The slide shows the hemoglobin molecule with iron binding O2.

Erythropoiesis feedback loop — memorize the chain

She asked: what is the PRIMARY stimulus for erythropoiesis?

Hypoxia (not enough O2 at the tissues) → kidney makes EPO → bone marrow makes RBCs → more O2 carrying capacity.

She said (not on the slide): *The diagram also showed the scale of production: on the order of 200 × 10^9 RBCs per day. Lungs pick up O2, tissues take it.*

TEST TIP: Hypoxia is the stimulus. The kidney is the sensor/EPO factory. The marrow is the factory floor. Do not skip a link.

Reticulocytes — almost-mature RBCs

RBCs leave the marrow slightly immature = reticulocytes. The developmental path on the slide: hemocytoblast → proerythroblast → … → reticulocyte → erythrocyte.

She said (not on the slide): *At any one time about 1% of circulating RBCs are reticulocytes. That is why we use a retic count as an index of marrow activity.*

Retic count	What it means	Possible reasons she gave
Lower than normal	Marrow is not making new RBCs as fast as usual	Missing building blocks (nutrition) OR marrow problem / suppression. Count does NOT tell you WHICH cause.
Higher than normal	Marrow is pumping out RBCs faster than usual (more still slightly immature)	Most common trigger she named: blood loss. Marrow tries to replace what was lost.

TEST TIP: Low retic = lazy or blocked marrow (or no ingredients). High retic = marrow is trying. After hemorrhage you EXPECT a high retic if the marrow works.

4. Blood types — ABO and Rh

ABO

RBCs have surface antigens used to group people. She walked the table out loud.

Type	Antigen on RBC	Antibody in plasma
A	A	Anti-B
B	B	Anti-A
AB	A and B	Neither
O	Neither	Anti-A and Anti-B

She said (not on the slide): *You are NOT born with these ABO antibodies. Babies spontaneously develop them by about 3–6 months of age WITHOUT being transfused the wrong blood — probably because similar antigens exist in the environment.*

Wrong-type transfusion: antigen meets its antibody → hemolysis (RBC bursting). That is why the bursting-cell slide sits right after ABO.

Rh

Named for the rhesus monkey (antigens discovered 1940). Many antigens (C, D, E); D is the most reactive and is used for Rh typing. Rh+ has the D/Rh antigen; Rh− does not.

DO NOT MIX UP: ABO antibodies appear spontaneously. Rh-negative people make anti-D ONLY if they are actually exposed to Rh-positive blood. Never exposed → never make the antibody.

If an Rh− person is exposed (wrong transfusion or an Rh− mother carrying an Rh+ fetus), they can then make anti-D. Later exposures can destroy Rh+ cells.

5. Polycythemia — too many RBCs

Polycythemia = too many red cells. Two predisposing factors on the slides, built one at a time.

1. Bone marrow disorder

Example: polycythemia vera — a type of blood cancer. Marrow itself overproduces RBCs (and often other lines).

2. Chronic hypoxia — she called this the MUCH more common reason

She said (not on the slide): *Hypoxia tells the kidney to make EPO. If you are always a little hypoxic, the kidney is always making extra EPO, so the marrow always makes extra RBCs.*

Causes of chronic hypoxia she listed:

- Smoking
- Chronic lung disease such as COPD (chronic obstructive pulmonary disease)
- Living at high altitude — less O2 in the air

She said (not on the slide): *Elite athletes exploit this: train at altitude → more RBCs → hope to carry more O2 to muscle → more endurance.*

Effects — two bullets, both testable

Effect	Why it matters
↑ Blood viscosity	Blood is thicker. Harder to push. Higher clot / sluggish-flow risk.
↑ Blood volume	More cells = more volume. Stresses the heart and vessels.

6. Anemia — not enough working RBCs / hemoglobin

Slide contrast: normal smear vs anemic smear (fewer cells). Anemia = lower than normal RBCs or hemoglobin, so tissues get less O2.

How patients feel / look (symptom slide)

Fatigue, weakness, dizzy, pale skin, rapid heartbeat, shortness of breath, chest pain, headache, leg pain, insomnia. These are the O2-delivery failure picture — not a separate disease.

Four predisposing factors — she built the list across many slides

Memorize all four. Questions often give a scenario and want the CATEGORY.

#	Category	High-yield examples from the recording
1	Nutritional deficiencies	Iron; B12 (cobalamin); B9 (folic acid)
2	Bone marrow failure	Aplastic anemia — empty / fatty marrow, not making cells
3	Blood loss	GI bleed, trauma, postop hemorrhage
4	Excessive / premature destruction (hemolysis)	Sickle cell (from last unit); toxins; burns; mechanical valves; autoimmune attack on RBCs

7. Nutritional anemias and the CBC

Iron deficiency → microcytic, hypochromic

Slide pair: normal peripheral smear vs microcytic hypochromic anemia. Cells are small (microcytic) and pale (hypochromic) because they lack hemoglobin/iron.

How you know without looking in a microscope — CBC

She said (not on the slide): *You look at MCV and MCHC on the CBC, not just the hemoglobin.*

Index	Stands for	What it tells you	In microcytic hypochromic iron deficiency
MCV	Mean corpuscular volume	SIZE of the RBC	DOWN (smaller cell → less volume)
MCHC	Mean corpuscular hemoglobin concentration	How much Hb is packed in each cell	DOWN (less Hb → lower concentration)

TEST TIP: Low MCV + low MCHC = microcytic hypochromic picture. She walked it as a Socratic question: if the cell is smaller, does volume go up or down? Down.

She said (not on the slide): *CBC also reports hemoglobin, RBC count, platelets, WBCs — but MCV/MCHC are the ones that classify the SIZE and COLOR of the anemia.*

B12 (cobalamin) and B9 (folic acid) → macrocytic / megaloblastic

These vitamins are needed to build DNA in rapidly dividing marrow cells. Without them, RBCs come out LARGE (macrocytic) and often megaloblastic (big, immature-looking). Retic path slide is reused to remind you where maturation can stall.

Vitamin	Also called	Classic extra problem besides anemia
B12	Cobalamin	Neuro changes (myelin)
B9	Folic acid / folate	Neural-tube defect / spina bifida in the fetus

B12 absorption and pernicious anemia

B12 needs intrinsic factor (IF) to get from gut into blood. IF is made by parietal cells in gastric mucosa.

She said (not on the slide): *B12 is a delicate vitamin. Digestive enzymes in the GI tract can destroy it. IF protects B12 until it can be absorbed.*

Ways she listed to lose IF:

- Chronic gastric inflammation / gastritis — stomach not making enough IF

- Part of the stomach removed — cancer surgery or weight-loss surgery — parietal cells gone
- Certain drugs interfering with production or activity of IF

Pernicious anemia = anemia because there is not enough IF, so B12 cannot be absorbed. She was very specific: ONLY use that name when IF is the reason B12 is missing.

TEST TIP: B12 deficiency from vegan diet or from ileal disease is still B12-deficiency anemia. It is pernicious anemia only when IF is the problem.

B12 deficiency — neuro changes

She said (not on the slide): *B12 is also needed to make normal myelin — the sheath around axons that speeds conduction. No normal myelin → neuro findings.*

Term	What she said it means
Paresthesias	Numbness, tingling, odd sensations
Ataxia	Coordination problems
Position sense / vibratory sense changes	Sensory deficits — you cannot tell where a limb is in space as well

TEST TIP: Folate deficiency does NOT give this neuro package. That contrast is high-yield even if she did not put it as a bullet.

B9 deficiency — spina bifida

Especially worried about pregnant women. Vertebrae do not close in the back → opening. Spinal nerves or even cord can protrude.

She said (not on the slide): *Degrees of severity on the picture: (1) bony defect only, nothing sticking out; (2) meninges ± spinal nerves sticking out; (3) actual cord sticking out. If nerves/cord are outside, they can be damaged → paralysis.*

Practical point she hit: make sure pregnant patients get their supplements.

Cheilosis / cheilitis

Cracked, inflamed corners of the mouth. Shown as a clinical clue that can travel with B-vitamin deficiency anemias.

8. Bone marrow failure

Aplastic anemia slide: marrow not producing the cell lines. Red cells (and often WBC + platelets) drop. Causes later overlap with thrombocytopenia and neutropenia: radiation, toxic drugs/chemicals, marrow cancer.

9. Blood-loss anemia

Acute bleed picture on the GI-bleed slide: pumping, oozing, exposed vessel, red or black clot.

Acute blood loss cells look normocytic normochromic — normal size, normal color — because you lost whole blood. You did not lose the ability to build hemoglobin inside each cell. MCV and MCHC stay normal at first.

TEST TIP: Chronic slow GI bleed can later look iron-deficient (microcytic hypochromic) once iron stores are spent. Acute hemorrhage starts normocytic.

Hypovolemia → shock — longest spoken segment in the lecture

Especially with a LARGE loss in a SHORT time, the problem is not only anemia. It is low circulating volume.

She said (not on the slide): *Word breakdown: hypo = low, vol = volume, -emia = blood → low blood volume.*

Why postop / post-procedure vital signs are frequent (she gave a schedule): maybe every 15 minutes the first hour, then every 30 minutes the next hour — to catch bleeding.

Physiology she taught	What you OBSERVE first
Lost volume → BP wants to fall	Heart rate rises FIRST (compensation: pump remaining blood into arteries faster)
Tachycardia can keep BP normal for a while	BP still normal + climbing HR = already worry
When HR can no longer compensate	THEN you see the BP drop
BP so low organs are not perfused	Shock → multi-organ failure → can die

She said (not on the slide): *Simplistic definition for this course (detail later in critical care): shock = BP has dropped to the point that you can no longer adequately perfuse major organs and they start to fail.*

TEST TIP: Patho logic: volume down → BP down. Bedside observation: tachycardia before hypotension. If you wait for the BP to fall, you are late.

10. Hemolytic anemia

Fourth cause: excessive or premature RBC destruction.

She said (not on the slide): *Callback to last class: sickle cell — sickled cells look defective so the body destroys them = hemolytic anemia.*

Other destroyers she listed:

- Toxic chemicals or bacterial toxins
- Physical trauma — severe burn damages RBCs in that area
- Artificial heart valves — even flexible valves can smash RBCs when they close
- Autoimmune disease — immune system attacks your own RBCs

Breakdown of hemoglobin → bilirubin → jaundice (yellow). That is why jaundice slides sit next to hemolysis and bile-path diagrams. Extra bilirubin dumped into the bile/liver pathway.

11. Hemostasis — how bleeding stops

She said (not on the slide): *She said her study guide lists FIVE stages of clot formation. She went through them in order.*

Stage	What happens	Triggered by / purpose
1. Vessel spasm /	Injured vessel clamps down	Injury + prostaglandins (e.g. thromboxane A2)

vasoconstriction		from injured cells. Limits blood leaving the hole.
2. Platelet plug (primary hemostasis)	Platelets stick to wall and to each other	Contact with injured wall + prostaglandins. Plugs the hole but is UNSTABLE — flow can knock it loose.
3. Fibrin clot (secondary hemostasis)	Coagulation cascade lays fibrin over the plug	Stabilizes the plug so it stays put.
4. Clot stays while vessel heals	Fibrin mesh holds	Cascade end-product is fibrin, not the platelet itself.
5. Clot degradation	Plasmin breaks fibrin down	Checks and balances — do not leave clots forever.

DO NOT MIX UP: Platelets ≠ clotting factors. Platelets = marrow, make the plug. Clotting factors = LIVER, make the fibrin clot that stabilizes the plug. She repeated this later and told you not to mix them up.

Intrinsic vs extrinsic cascade

Pathway	Activated by	Spoken example
Intrinsic	Injury to the BLOOD VESSEL	Atherosclerosis can injure the vessel without injuring surrounding tissue — so the two paths are not totally redundant
Extrinsic	Injury to TISSUES	Cut, trauma — tissue factor

She said (not on the slide): *She used to think two pathways were redundant (how do you injure a vessel without injuring tissue?). Atherosclerosis is the exception. Newer research: more overlap than textbooks first taught — still teach both until they know more.*

Both pathways start on different factors, then JOIN. End result of either path: fibrin clot. Factors circulate in INACTIVE form and sequentially activate the next one (cascade).

She said (not on the slide): *Checks and balances: blood also has anticoagulant / anti-clotting factors so you only clot where you need to, not everywhere.*

12. Platelet problems — thrombocytopenia

Example she used: thrombocytopenia = too few platelets. Bleeding because you cannot build a stable plug.

Predisposing factors (built across slides)

Factor	Why
Bone marrow depression	Marrow makes platelets. Radiation, toxic drugs/chemicals, marrow cancer → fewer platelets out.
Splenomegaly	Spleen = large. After release, platelets hang in the spleen up to ~8 hours. Big spleen sequesters more platelets AND over-clears old/defective cells, including platelets.
Autoimmune destruction	Antibodies attack platelets (ITP-type picture).

What bleeding looks like

- Epistaxis = nosebleed (slide title).

- Petechiae = pinpoint bleeds in skin (photo slides).

Easy bruising in general when plugs cannot form.

Interventions she named when the cause is autoimmune

- Splenectomy — take out the organ that is sequestering / eating platelets.
- Immunosuppressants — quiet the immune attack.
- Plasmapheresis / plasma exchange — machine looks like dialysis. Pull blood, separate plasma from cells, give cells back with synthetic plasma, throw away the patient's plasma because that is where the attacking antibodies live. Two or three rounds can put some patients into remission. Not every autoimmune disease responds.

Platelets that exist but do not WORK — antiplatelet drugs

She said (not on the slide): *Before leaving platelets she added a whole spoken section that is barely a title on the next slide: people can bleed with a NORMAL platelet COUNT if the platelets cannot clump.*

Aspirin is the prototype antiplatelet. It stops platelets from clumping into a plug → bleed longer, bruise more.

Same property is used ON PURPOSE: prevent arterial clots that cause ischemic stroke / TIA / stable angina / ACS / PAD / MI.

She said (not on the slide): *Mini-dose / baby aspirin once a day, or one adult aspirin every other day, is NOT for headaches or cramps. One baby aspirin will not treat pain. If you see that dose, it is for antiplatelet effect.*

Slide table of antiplatelet classes (know they exist and that bleeding is the shared adverse effect):

- Aspirin — GI bleed, hemorrhagic stroke risk.
- P2Y12 blockers: clopidogrel (Plavix), prasugrel (Effient), ticlopidine (Ticlid) — ACS / stent / stroke / PAD; Ticlid also thrombocytopenia, neutropenia, TTP.
- PDE inhibitors: dipyridamole (Persantine) after valve replacement with warfarin; cilostazol (Pletal) for claudication.
- GP IIb/IIIa inhibitors (ReoPro, Integrilin, Aggrastat) — potent; GI / urinary / retroperitoneal bleeding.

13. Clotting-factor problems

Two directions on the slide: too many factors / too much clotting, or not enough.

Too many / too much clotting → thromboembolic events

Thrombus = clot where it formed. Embolus = clot that traveled. Together: thromboembolic events (MI, ischemic stroke, PE, DVT picture).

Not enough factors — four predisposing factors

Factor	Mechanism she taught
Liver disease	Liver MAKES clotting factors. Sick liver → not enough factors. Also may not make bile.

Vitamin K deficiency	Many factors need vit K to be made in a WORKING form. Factors may be produced but they do not function.
Hereditary disorders	Classic: hemophilia (Factor VIII was the example in the genetics unit).
Increased consumption	Used up faster than replaced — lead-in to DIC.

How you become vitamin K deficient (spoken, not all on the slide):

- Diet low in vit-K foods.
- Cannot ABSORB it: vit K is fat-soluble → needs BILE. Liver disease → no bile → no vit K absorption → now you fail to make working factors TWO ways (liver cannot synthesize them AND cannot absorb K).
- Broad-spectrum antibiotics (BSA) — the slide after vit K. Gut bacteria help make vit K. Wipe them out with BSA → less K → worse clotting.

TEST TIP: Platelets = marrow. Factors = liver + vitamin K. BSA hits the bacterial K source, not the marrow.

DIC — disseminated intravascular coagulation

She said (not on the slide): *Break the words: coagulation = clotting; intravascular = inside vessels; disseminated = widespread. So: clotting in vessels all over the body.*

NOT a primary disease. It is a complication of another critical illness: severe infection, multiple trauma, advanced cancer — ICU patients. Inflammation / chemical mediators of inflammation are thought to flip the cascades on.

Two problems at once:

- Microthrombi occlude flow → tissue ischemia → organ failure.
- Making all those clots CONSUMES the clotting factors → nothing left when you need to stop a bleed → hemorrhage.

She said (not on the slide): *Paradox: abnormal clotting AND abnormal bleeding at the same time. Some patients show only one side.*

TEST TIP: DIC is the example of “increased consumption” on the factor-deficiency list.

14. White blood cells

She said (not on the slide): *She got confused in nursing school because one cell can have several names. She tried to collapse the aliases. There are not as many types as it feels.*

Three MAIN groups: granulocytes, monocytes, lymphocytes.

Granulocytes — granules visible under the microscope

Cell	Also called	Job from the recording
Neutrophil	Polymorphonuclear leukocyte / PMN; mature = segs; immature = bands	Phagocyte. First-line eater of bacteria / foreign material. Lysosomal enzymes digest what it engulfs.
Basophil	A granulocyte	She grouped it with the other two granulocytes on the lineage slide (allergy / histamine role is the usual pairing — lecture emphasis was the trio).
Eosinophil	A granulocyte	Primary job: parasites (too big to phagocytose). Granule enzymes damage parasites. Also pile up in

		allergy / asthma. Some enzymes pro-inflammatory, some anti-inflammatory — likely different subtypes. Body's checks and balances.

Segs vs bands and "shift to the left"

CBC often splits neutrophils into segs and bands.

- Segs = MATURE neutrophils. Nucleus looks segmented.
- Bands = IMMATURE neutrophils. Nucleus is a horseshoe / band.

She said (not on the slide): *Why "left"? Old lab slips listed bands on the LEFT and segs on the RIGHT. More bands = shift toward the left column. Modern slips stack them top/bottom but the phrase stuck.*

Shift to the left = more bands than usual. Most common cause she named: infection (or inflammation). Marrow speeds neutrophil production so more slightly immature cells spill out. Also called neutrophilia (too many neutrophils).

TEST TIP: Left shift on a CBC is a clue the marrow is responding to infection, not a random lab quirk.

Monocytes and lymphocytes (preview of the immune lecture)

- Monocytes: phagocytes in BLOOD. When they leave the vessel into tissue they are called macrophages or dendritic cells. More next class.
- Lymphocytes — adaptive immunity, more next class:
- B lymphocytes mature in Bone marrow → become antibody producers → humoral immunity.
- T lymphocytes mature in Thymus → cell-mediated immunity.
- NK (natural killer) cells — innate immunity.

Too few WBCs — the pinea words

She said (not on the slide): *Suffix -penia = deficiency. Prefix a- = absence / nearly without.*

Term	Means	How worried
Leukopenia	Too few leukocytes (WBCs in general)	General term
Neutropenia	Too few neutrophils	High concern — neutrophils are the most plentiful WBC and first phagocytes on scene
Granulocytopenia	Too few granulocytes (often used like neutropenia)	Same neighborhood
Agranulocytosis	Neutrophils so low it is almost like having none	She used a number: neutrophil count < 200. Normal frame she gave: about 3,000–7,000. Below 200 ≈ without granulocytes

Neutropenia — predisposing factors (same logic as platelets)

Factor	Spoken explanation
Bone marrow problems	Marrow makes neutrophils. Same insults: radiation, drugs, chemicals, marrow cancer.
Increased destruction (autoimmune)	Immune system attacks neutrophils.
Splenomegaly	Spleen clears old/defective blood cells. Big spleen over-clears neutrophils too.

Increased consumption	Used up faster than replaced. Neutrophils migrate into a bad wound/infection and leave the bloodstream. A raging infection can drain circulating neutrophils faster than marrow can refill.

Assessment / interventions slide was a prompt for you to think: infection risk, fever, mucous membranes, protective precautions — and drugs that stimulate production.

G-CSF (granulocyte colony-stimulating factor) — the synthetic growth factor she flagged at the start of the lecture. Give it to push the marrow to make neutrophils.

15. Review: cancer applied to blood — leukemia

She rebuilt the cancer list from the previous unit, one bullet per slide, then applied it to WBC cancers.

Cancer rule from last unit	How it shows up in leukemia
Cells don't function properly	Blasts / poorly differentiated WBCs cannot do mature immune jobs
Cells crowd out normal cells	Marrow packed with blasts → anemia + thrombocytopenia + neutropenia together
Can metastasize	Leukemic cells already live in blood; infiltrate organs
Causes anorexia / cachexia	Same two mechanisms as last unit: steal nutrients + cytokine anorexia
Causes pain	Marrow expansion, infiltration, ischemia, tissue damage

Blasts

Very immature white-cell precursors. Acute leukemias are blast-heavy. Poorly differentiated — callback to the differentiation ladder from Cell Part 2.

Acute vs chronic

- Acute: sudden, many blasts, poorly differentiated, sick fast.
- Chronic: more mature-appearing cells, slower course (still cancer).

Healthy vs leukemia marrow diagram: healthy side has stem cells → fully differentiated immune cells. Leukemia side is crowded with poorly differentiated cells.

How we look at the marrow

She said (not on the slide): *Aspiration = withdraw fluid. Biopsy = bigger needle, actual chunks of marrow. Usual site: iliac crest. Alternatives: tibia; sternum if those cannot be used. Sternum looks scary — periosteum is tough; they push hard. She watched one at the bedside.*

Goal of the specimen: see how differentiated the cells are, stage the leukemia, name the specific type.

Stem-cell / bone-marrow transplant (process slide)

Collection from patient or donor blood/marrow → lab processing to purify/concentrate → cryopreservation (freeze) → patient gets high-dose chemo and/or radiation to wipe marrow → thawed cells reinfused. Allogeneic = donor cells.

16. Rapid-fire test bank

- Marrow makes erythrocytes, leukocytes, thrombocytes.

- Hematopoiesis is growth-factor driven (EPO, G-CSF); we give synthetic versions.
- RBC job = O2 via iron in hemoglobin.
- Erythropoiesis chain: hypoxia → kidney EPO → marrow RBCs.
- Retic ~1% of RBCs. Low retic = sluggish marrow or no ingredients. High retic = marrow responding; most common trigger = blood loss.
- ABO antibodies by 3–6 months without transfusion. Rh antibodies only after exposure.
- Polycythemia: PV (marrow cancer) vs secondary from chronic hypoxia (smoke, COPD, altitude, athletes). Effects = viscosity + volume.
- Anemia 4 causes: nutrition, marrow failure, blood loss, hemolysis.
- Iron: microcytic hypochromic; MCV↓ MCHC↓.
- B12/B9: macrocytic/megaloblastic. Pernicious = no IF (parietal cells / gastritis / gastrectomy / some drugs). B12 neuro = myelin (paresthesia, ataxia, position/vibration). B9 = spina bifida.
- Acute bleed: normocytic normochromic. Watch HR first, then BP. Shock = organs not perfused.
- Hemolysis: sickle, toxins, burns, mechanical valves, autoimmune → jaundice.
- Hemostasis: spasm → platelet plug (unstable) → fibrin clot (cascade) → plasmin later. Intrinsic = vessel; extrinsic = tissue; they join.
- Platelets ≠ factors. Thrombocytopenia: marrow, big spleen (8-hour hangout + over-clearing), autoimmune. Treat autoimmunity with splenectomy, immunosuppressants, plasmapheresis.
- Aspirin / antiplatelets: bleed more; used to prevent arterial clots. Baby aspirin = antiplatelet dose, not analgesia.
- Factor deficiency: liver, vit K (diet, no bile, BSA), hereditary, consumption/DIC.
- DIC: ICU complication; microclots + used-up factors = clot AND bleed.
- Neutrophil = PMN = segs (mature) / bands (immature). Left shift = infection. Eosinophils = parasites + allergy. Mono in blood = macrophage/dendritic in tissue. B/T/NK preview.
- Leukopenia vs neutropenia vs granulocytopenia vs agranulocytosis (<200 neutrophils vs normal ~3,000–7,000).
- Neutropenia causes: marrow, autoimmune destruction, spleen, consumption into infected tissue. Drug: G-CSF.
- Leukemia = blood-line cancer using the five cancer rules. Blasts. Aspirate vs biopsy. Transplant steps.

17. End-of-deck cartoons (not test content)

Slides 94–105 are cartoons (leukocyte / platelet / iron / pernicious-definition jokes) and a “doctor wants more blood tests” closer. Use them as a mental break, not as sources of new mechanisms.

Pernicious on those last slides is the dictionary joke (destructive / deadly) — a memory hook for why they named IF-related B12 anemia “pernicious.”